



Lecture 23:

VLSI Jitter Amplification and Jitter Filtering Effects

Contents



This Topic: VLSI Jitter/Jitter Amplification, and Jitter Filtering

- Introduction
- Jitter/Jitter Transfer Function
- Jitter Amplification Effects
- Jitter Filtering Effects

Introduction



- Jitter may interact with the VLSI circuits
- Jitter will experience amplification or attenuation effects
- Impacts of I/O circuits to jitter can be modeled using Jitter/Jitter (JJ) transfer function

Circuit Types based on Jitter TF



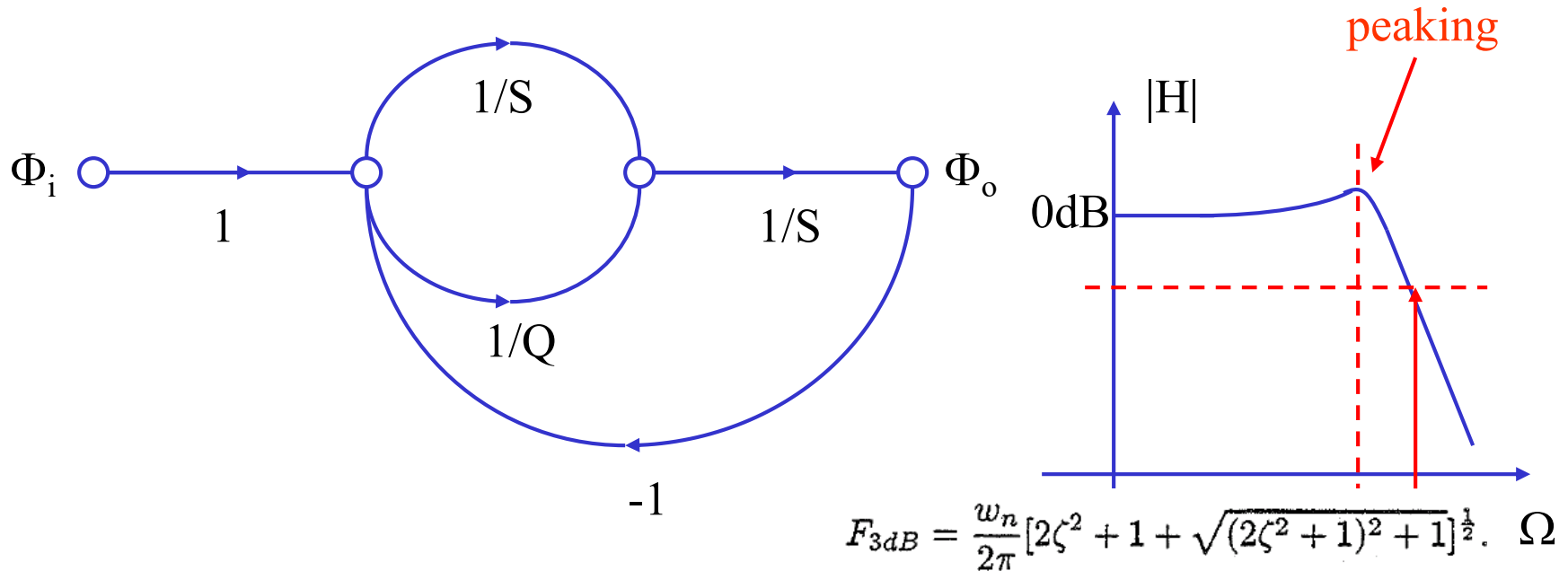
- Two basic circuit families which impact jitter in different ways
 1. Phase Domain Circuits
 2. Filter-Like Circuits
- For phase domain circuit (e.g. PLL, DLL, DRC, etc.), the jitter transfer function is basically the transfer function of the circuit
- For filter-like circuit (e.g. Tx, Rx, Buffer, Channel, Filter, etc.), jitter transfer function is different from signal transfer function of the circuit

Jitter TF of Phase Domain Circuits



- For phase domain circuit, design parameters are the input and output clock/signal phase.
- The frequency domain input/output phase transfer function directly provide the jitter transfer function ($H(s) = JJ(s)$)
- This type of circuits are commonly used for jitter spectral shaping and filtering
- Detail analysis of jitter transfer function has been provided in earlier lectures

Example: Jitter TF of 2nd Order PLL



$$JJ(s) \equiv H(s) \equiv \frac{\phi_o}{\phi_i} = \frac{\frac{1}{Q}S + 1}{S^2 + \frac{1}{Q}S + 1}$$

$$|H(s)|_{s=j\omega} = \begin{cases} 1 & \omega = 0 \\ \sqrt{1 + Q^2} > 1 & \omega = 1 \\ 0 & \omega = \infty \end{cases}$$

Jitter TF of Filter-Like Circuits

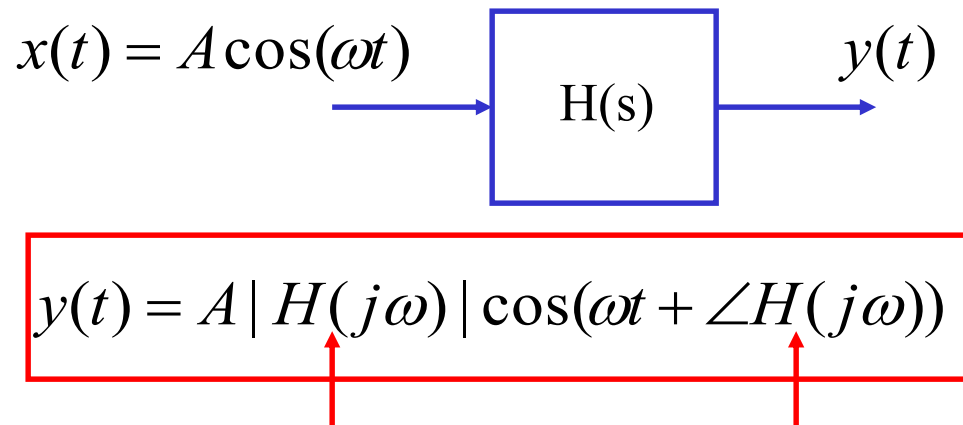


- For filter-like circuit, design parameters are the input and output voltage (current). The frequency domain input/output transfer function of the signal $H(s)$ is different from its jitter transfer function $JJ(s)$
- This type of circuits are commonly used for voltage (current) signal spectral shaping and filtering
- In general it can be modeled using a linear filter

Impact of Linear Filter to Clock Signals



- Filter may impact both clock swing and phase shift (i.e. Bode Plots)
- For ideal clock, no dispersion occur and there is only a static phase shift (skew)
- No jitter injection occurs



Gain Response

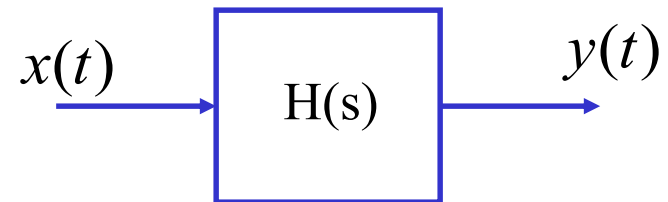
Phase Response

S-domain Model of Jitter Transfer Function in Linear Filter



- Signal transfer function

$$H(s) \equiv \frac{A_o(s)}{A_i}$$



- Jitter Transfer Function

$$JJ(s, \omega_m) \equiv \frac{\beta_o(s, \omega_m)}{\beta_i}$$

$$x(t) = A \cos(\omega t + \beta_i \cos(\omega_m t + \phi_i))$$

Jitter Magnitude

Jitter Frequency (tone)

$$y(t) = A_o \cos(\omega t + \beta_o \cos(\omega_m t + \phi_o))$$

Small Signal Jitter Modeling



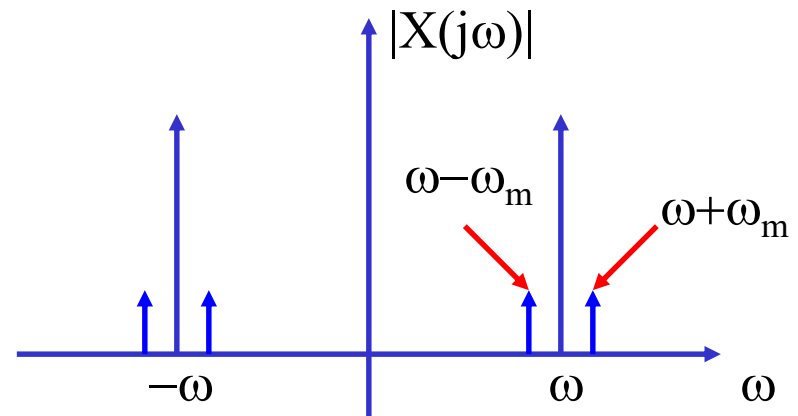
- For single-tone small jitter in sinusoidal clock signal:

$$x(t) \equiv \text{Re}[x(t)^*]$$

$$x(t)^* = e^{j(\omega t + \beta_i \cos(\omega_m t))} \approx (1 + j\beta_i \cos(\omega_m t))e^{j\omega t}$$

$$= (1 + \frac{\beta_i}{2} (e^{j\omega_m t} + e^{-j\omega_m t}))e^{j\omega t}$$

$$= e^{j\omega t} + \frac{\beta_i}{2} (e^{j(\omega + \omega_m)t} + e^{j(\omega - \omega_m)t})$$



Note:

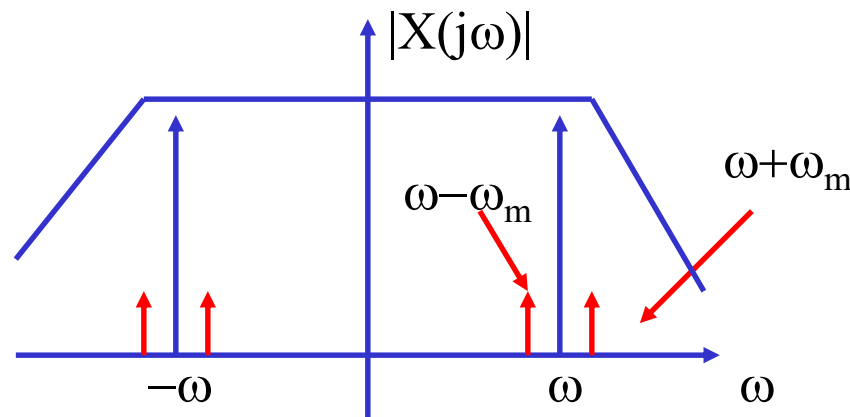
- Jitter can be modeled as a phase modulation
- Under such model, jitter is proportional to the magnitude ratio of the jitter tone to the carrier (i.e. ideal clock)

Jitter Transfer Function of Linear Filter



- Steady-State Response of a filter-like circuit:

$$\begin{aligned}
 y(t)^* &= H(j\omega)e^{j\omega t} + \frac{\beta_i}{2} (H(j(\omega + \omega_m))e^{j(\omega + \omega_m)t} + H(j(\omega - \omega_m))e^{j(\omega - \omega_m)t}) \\
 &= H(j\omega)e^{j\omega t} \left[1 + \frac{\beta_i}{2} \left(\frac{H(j(\omega + \omega_m))}{H(j\omega)} e^{j\omega_m t} + \frac{H(j(\omega - \omega_m))}{H(j\omega)} e^{-j\omega_m t} \right) \right]
 \end{aligned}$$



Note: JJ is related to the ratio of the TFs of the jitter tones to the carrier

Case I: Wideband Circuits



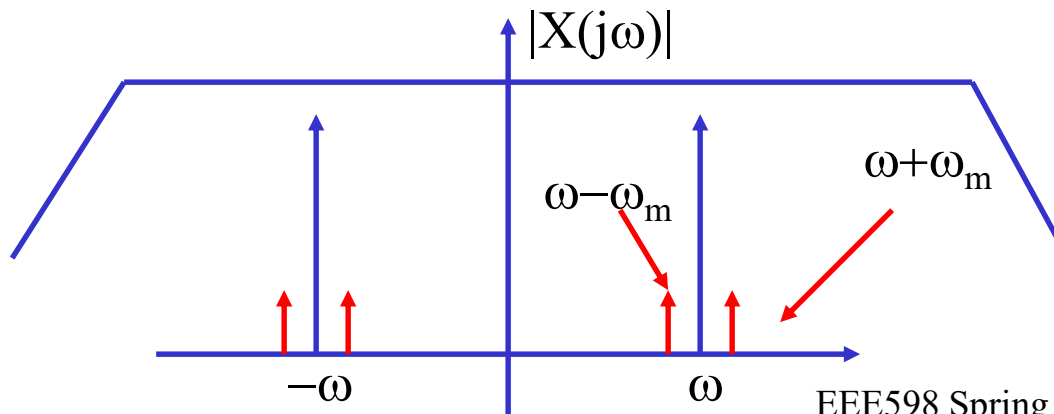
- For wideband circuits ($BW \gg f_{ck}$)

$$H(j\omega) \approx H(j(\omega + \omega_m)) \approx H(j(\omega - \omega_m))$$

$$y(t)^* = H(j\omega)e^{j\omega t} \left[1 + \frac{\beta_i}{2} (e^{j\omega_m t} + e^{-j\omega_m t}) \right]$$

$$JJ(j\omega, \omega_m) \approx 1$$

- Jitter is not impacted



Case II: Band-Limited Circuits

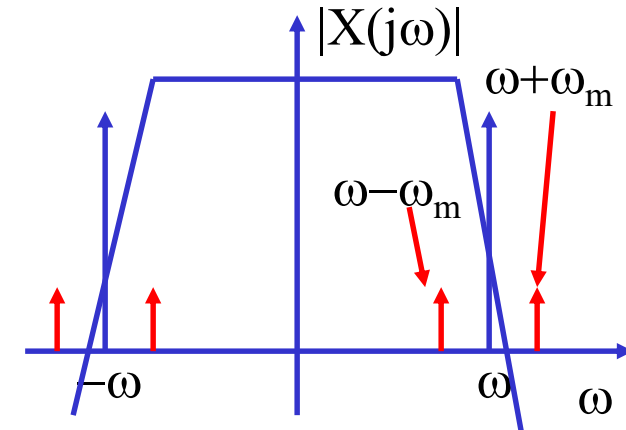


- When circuit is Band-Limited

$$|H(j(\omega - \omega_m))| \gg |H(j\omega)| \gg |H(j(\omega + \omega_m))|$$

$$y(t)^* \approx H(j\omega)e^{j\omega t} \left[1 + \frac{\beta_i}{2} \left(-\frac{H(j(\omega - \omega_m))}{H(j\omega)} e^{-j(\omega_m t + \phi)} \right) \right]$$

$$\approx H(j\omega)e^{-\frac{\beta_i}{2} \left| \frac{H(j(\omega - \omega_m))}{H(j\omega)} \right| \cos(\omega_m t + \phi)} e^{j\left(\omega t + \frac{\beta_i}{2} \left| \frac{H(j(\omega - \omega_m))}{H(j\omega)} \right| \sin(\omega_m t + \phi)\right)}$$



Amplitude Modulation effect

- Jitter will be amplified

$$JJ(j\omega, \omega_m) \approx \frac{H(j(\omega - \omega_m))}{2H(j\omega)}$$

$$|JJ(j\omega, \omega_m)| \approx \left| \frac{H(j(\omega - \omega_m))}{2H(j\omega)} \right| \gg 1$$

Jitter Amplification effect

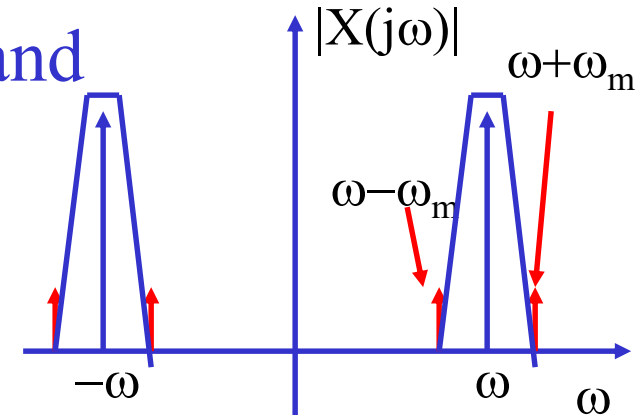
Case III: Narrow Bandpass Circuits



- When carrier is at center of passband

$$|H(j\omega)| \gg |H(j(\omega + \omega_m))|$$

$$|H(j\omega)| \gg |H(j(\omega - \omega_m))|$$



$$y(t)^* = H(j\omega)e^{j\omega t} \left[1 + \frac{\beta_i}{2} \left(\frac{H(j(\omega + \omega_m))}{H(j\omega)} \right) e^{j\omega_m t} - \frac{H(j(\omega - \omega_m))}{H(j\omega)} e^{-j\omega_m t} \right]$$

- Jitter will be reduced

$$JJ(j\omega, \omega_m) \sim \frac{H(j(\omega - \omega_m)) + H(j(\omega + \omega_m))}{2H(j\omega)}$$

$$|JJ(j\omega, \omega_m)| \ll 1$$

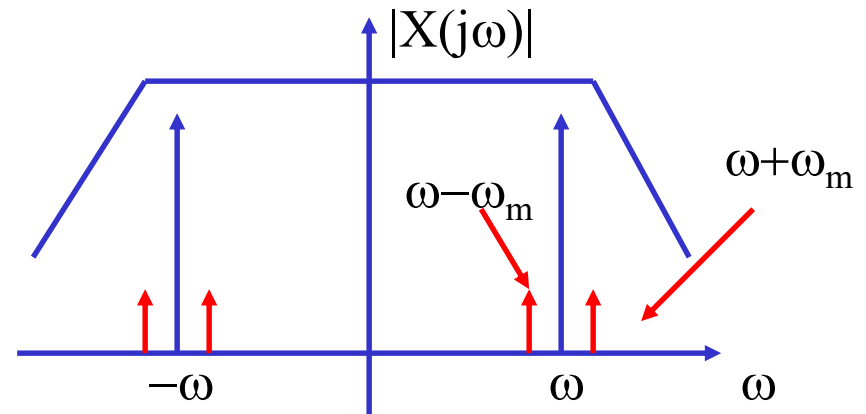
Jitter is attenuated
(i.e. jitter Filtering Effects)

Case Study: JJ of 1st Order Filter



- For 1st-Order Lowpass Filter

$$H(j\omega) = \frac{1}{1 + j\left(\frac{\omega}{\omega_o}\right)}$$



$$y(t)^* = H(j\omega)e^{j\omega t} \left[1 + \frac{\beta_i}{2} \left(\frac{(1 + j\left(\frac{\omega}{\omega_o}\right))e^{j\omega_m t}}{1 + j\left(\frac{\omega + \omega_m}{\omega_o}\right)} - \frac{(1 + j\left(\frac{\omega}{\omega_o}\right))e^{-j\omega_m t}}{1 + j\left(\frac{\omega - \omega_m}{\omega_o}\right)} \right) \right]$$

Case Study: JJ of 1st Order Filter



- For in-band clocking $\omega_o \gg \omega + \omega_m$

$$\begin{aligned} y(t)^* &\approx H(j\omega)e^{j\omega t} \left[1 + \frac{\beta_i}{2} \left(\left(1 - j \left(\frac{\omega_m}{\omega_o} \right) \right) e^{j\omega_m t} - \left(1 + j \left(\frac{\omega_m}{\omega_o} \right) \right) e^{-j\omega_m t} \right) \right] \\ &= H(j\omega)e^{j\omega t} \left[1 + \frac{\beta_i}{2} \sqrt{1 + \left(\frac{\omega_m}{\omega_o} \right)^2} (e^{j(\omega_m t - \varphi)} - e^{-j(\omega_m t - \varphi)}) \right] \end{aligned}$$

$$|JJ(j\omega, \omega_m)| \approx \sqrt{1 + \left(\frac{\omega_m}{\omega_o} \right)^2} \sim 1$$

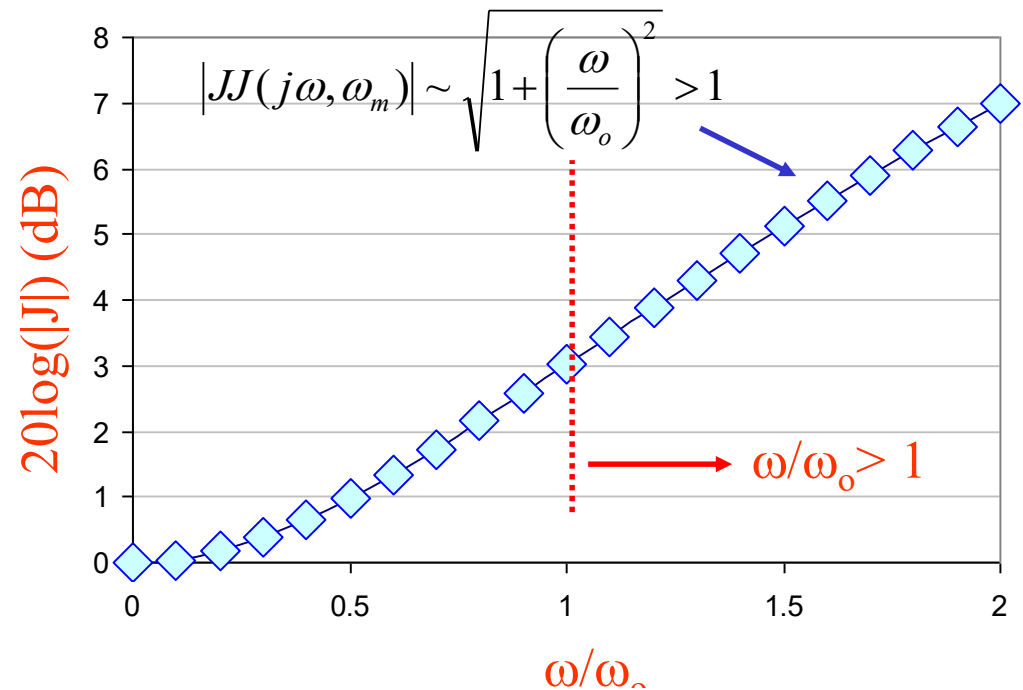
Case Study: JJ of 1st Order Filter

- For band-limited clocking circuit $\omega_o \ll \omega$, $\omega_m \sim$

ω

$$y(t)^* \approx H(j\omega)e^{j\omega t} \left[1 - \frac{\beta_i}{2} \left(1 + j \left(\frac{\omega}{\omega_o} \right) \right) e^{-j\omega_m t} \right]$$

$$|JJ(j\omega, \omega_m)| \sim \sqrt{1 + \left(\frac{\omega}{\omega_o} \right)^2} > 1$$



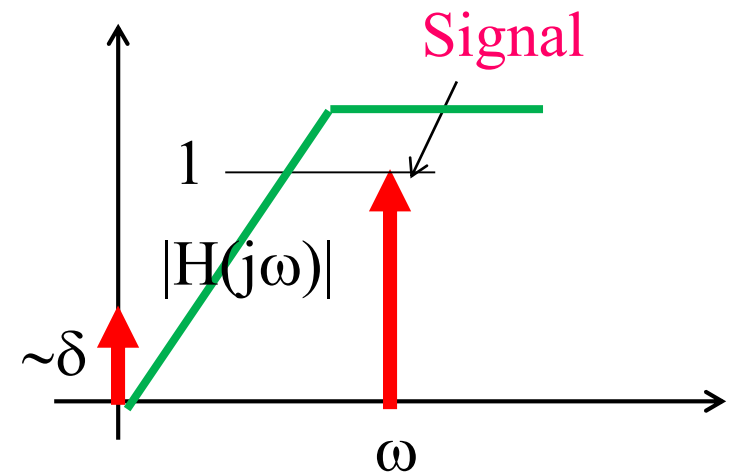
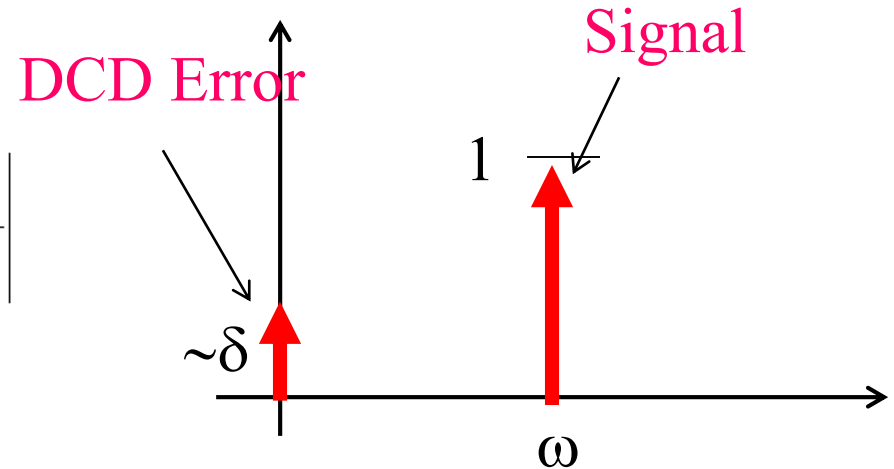
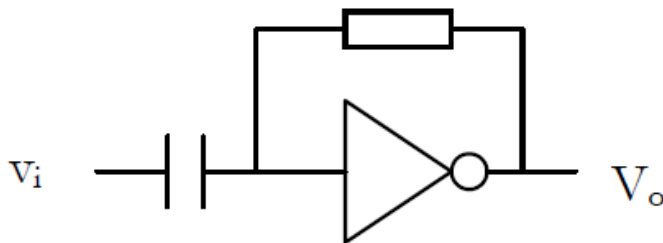
Application: DCD Correction



- DCD Model

$$DCD \equiv \frac{t_{High_Side} - t_{Low_Side}}{2T} \approx \frac{1}{2\pi} \left| \frac{V_{dc}}{V_{ac}} \right|$$

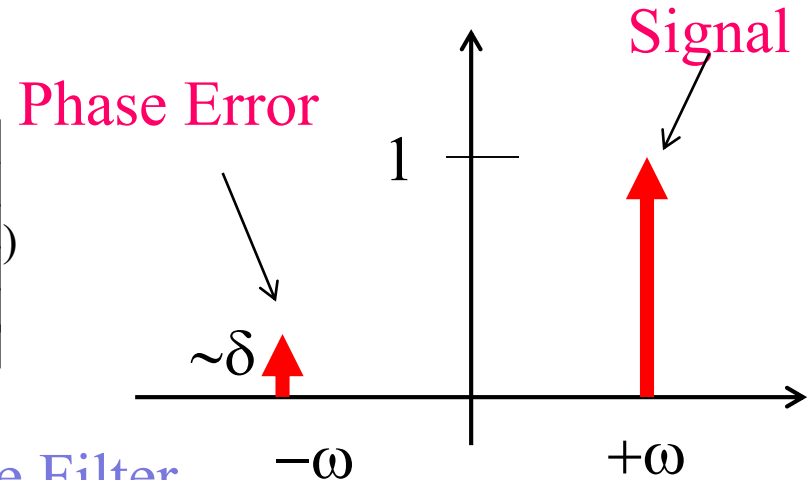
- DCD Filter



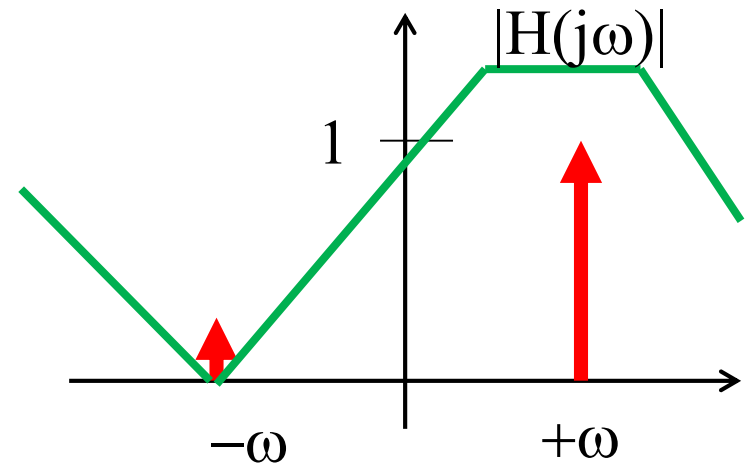
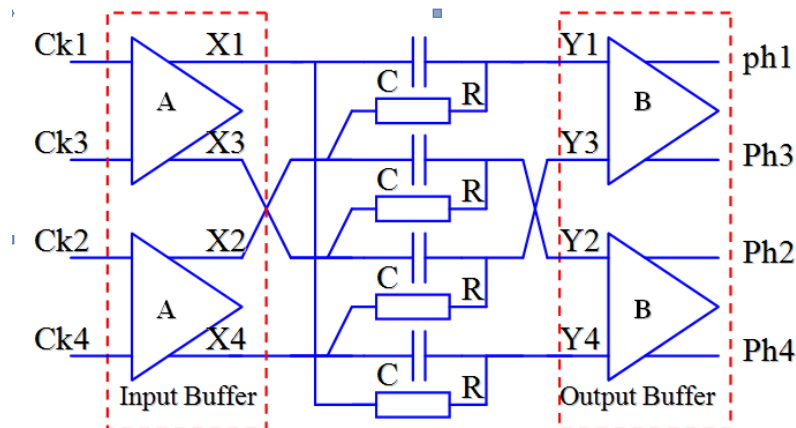
I/Q Phase Error Filter

- I/Q Clock Phase Error Model

$$\begin{pmatrix} ck_1 \\ ck_2 \\ ck_3 \\ ck_4 \end{pmatrix} = \text{Re}(e^{j\omega t} \begin{pmatrix} e^{j\delta/2} \\ e^{j\pi/2 - j\delta/2} \\ e^{j\pi + j\delta/2} \\ e^{j3\pi/2 - j\delta/2} \end{pmatrix}) \approx \text{Re}(e^{j\omega t} \left(\begin{bmatrix} 1 \\ j \\ -1 \\ -j \end{bmatrix} + \frac{\delta j}{2} \begin{bmatrix} 1 \\ -j \\ -1 \\ j \end{bmatrix} \right))$$



- I/Q Phase Error Filter: Polyphase Filter



Reading Assignment



- Please read Chapter 18 of the textbook